

Your program (PID 12345)

`ipc_range()` starts

creates `~/ .ipc/12345.fifo` (0600)

registers in `~/ .ipc/registry.json`

installs `SIGUSR1` handler

[loop running ...]

signal handler fires

reads `"peek\n"` from FIFO

prints status to stdout

[loop continues ...]

Your second terminal

(loop already running)

↓ user types `ipc peek` ↓

`$ ipc peek 12345`

opens `~/ .ipc/12345.fifo` for writing

writes `"peek\n"` (atomic, $\leq$ PIPE_BUF)

sends `SIGUSR1` to PID 12345

(returns / exits)

~/ .ipc/12345.fifo (shared)

SIGUSR1 delivered

Startup / initialisation

Loop execution

Signal handler

CLI (ipc command)